QUANTITATIVE METRIC WALL CROSSING FOR K3 BUNDLE MODULI

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Abstract. We compare harmonic L^2 metrics on bundle moduli over a projective K3 surface with flat quiver quotient metrics near a polystable bundle. Its locally free factors are slope-stable with a common reduced Hilbert polynomial for an ample polarization, and slope-stable of a common slope for a central Ricci-flat Kähler background, whose class may differ. The multiplicative metric error tends to zero uniformly as stabilizers change and positive degeneration scales separate arbitrarily. The proof retains all parallel automorphisms in the finite-dimensional reduction and computes harmonic lengths by complementary gauge minimization. For doubled quivers admitting a representation with no nonzero proper subrepresentation in the complex zero moment fiber, we prove pointed Gromov–Hausdorff continuity of the complete stratified quotients as real central parameters vary. Under this hypothesis, finite successive rescalings of the actual moduli metric have complete limits whenever the rescaled parameters and balanced normal base representatives converge at each stage. We apply the method to a rank-five K3 bundle with three spherical factors, multiplicities $(2, 2, 1)$, and two independent polarization parameters. Its five realized walls have selected transverse contractions with models $T^*\mathbb{P}^9$, $T^*\mathbb{P}^{11}$ and $T^*\mathbb{P}^{17}$. We determine their contraction scales, distances to the corresponding strata, and matching to the surrounding wall metric.

1. Introduction

Near a reducible polystable bundle, deformation theory gives a finite quotient description of the moduli germ. The harmonic L^2 metric contains further information. It is obtained by solving the Hermitian–Einstein equation and then taking harmonic representatives of deformations. When the stabilizer changes, the finite orbit operator develops small eigenvalues, so an inverse estimate that treats all gauge directions together is not uniform. This distinction is especially relevant when several stability parameters approach different walls at different rates.

We prove a local comparison between the actual harmonic metric and the flat quiver quotient metric that remains relative through these degenerations. The geometric setting is a projective K3 surface with a polystable locally free sheaf $E_0 = \bigoplus_i F_i \otimes \mathbb{C}^{m_i}$. The factors are pairwise nonisomorphic slope-stable bundles with equal reduced Hilbert polynomial for an integral ample class. The central analytic metric is Ricci-flat in a Kähler class for which the factors are stable of a common slope; the two classes may coincide. These hypotheses are verified directly in the application below, including at its selected noncentral wall points.

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The central harmonic norm and Serre duality identify $V = \mathrm{Ext}^1(E_0, E_0)$ with a unitary doubled-quiver representation. Its complex moment zero locus gives the convergent integrability model. For nearby Kähler forms ω_t, the real central parameter is the degree difference

$$\zeta_i(t) = 2\pi\big(\mathrm{rk}(F_i)\mu_{\omega_t}(E_0) - \deg_{\omega_t} F_i\big).$$

Theorem 3.1 identifies full local neighborhoods of actual HYM solutions with the corresponding finite quotients. On each genuine smooth-stratum tangent space, at a representative of harmonic norm r, it proves

$$f_t^* g_t^{L^2} = (1 + O(|t| + r))g_{\zeta(t)}$$

in the sense of two-sided inequalities of Hermitian forms. The constant does not depend on stabilizer type or ratios among positive finite orbit eigenvalues. A verified central sign symmetry improves r to r^2; no such cancellation is assumed otherwise.

1.1. **The analytic argument.** All parallel automorphisms are retained in the finite-dimensional problem. On their fixed infinite-dimensional orthogonal complement, the gauge equation has a uniformly coercive elliptic inverse. The remaining finite moment map is the moment of a smooth closed invariant two-form close to the flat form.

The decisive finite estimate is relative to each orthogonal vertex projection P:

$$|\langle \rho(x) - \mu(x), P\rangle| \le \varepsilon \|C_x P\|^2.$$

Here μ is the flat moment, ρ the perturbed normalized moment, and C_x the infinitesimal action. The same crossing energy appears in the positive moment increment across a spectral cut. Working in the Hermitian centralizer of the actual compact stabilizer, orthogonal to its infinitesimal stabilizer, makes this crossing energy strictly positive at every nonzero boundary correction. It cancels from the estimate. A degree argument gives a bounded complex-orbit correction and a reverse correction without assuming a uniform positive orbit gap.

For the metric, first minimize over the fixed complementary gauge directions. The resulting smooth finite Hermitian form has the flat harmonic norm as its value at the origin. Minimizing next over the retained finite algebra gives exactly the actual harmonic Kodaira–Spencer norm. Derivatives of the orbit correction are complex vertical and disappear in this infimum. This yields relative quotient estimates without bounding those derivatives. A separate stratum implicit-function argument and a full local inverse supply the differentiability and coverage needed in this calculation.

These steps develop methods from the preceding stability-degeneration papers. The spectral-cut calibration is from [Wue26a], and its repeated-factor form is developed in [Wue26c]. Relative moduli-metric comparison by complementary minimization is developed in [Wue26b, Wue26c]. The projection-relative moment estimate and the harmonic coordinate argument were used for the two-factor K3 example in [Wue26d]. Here the finite correction treats arbitrary quiver stabilizers, including loops, and its reverse direction covers the full nearby solution space. We give the required proofs rather than import conclusions restricted to those earlier representations.

1.2. **Intrinsic consequences.** A comparison of tangent norms alone does not determine intrinsic distances or a complete rescaled limit: short competing paths could leave the chart, and an image of a fixed model ball need not cover the corresponding actual ball. We establish both path localization and inverse coverage.

The finite-dimensional result is also useful independently. For any doubled quiver whose complex zero fiber contains a module-simple representation, meaning one with no nonzero proper subrepresentation, Theorem 5.1 proves that the full stratified length quotient at every

real central parameter is a proper geodesic space. Arbitrary converging real parameters give converging distances for converging balanced representatives, and convergence to the whole complete pointed quotient, including singular levels. The proof uses connected fibers of projective face maps to construct shrinking neighborhoods, uniform semialgebraic paths for short joins, and a quantitative Hamiltonian slice to retain asymptotically exact lengths along strata. Horizontal lifting gives the opposite distance inequality.

Combining this result with the harmonic comparison gives complete actual rescaled limits whenever $\zeta(t_n)/\delta_n^2$ converges. At a moving wall point, the full normal Ext-quiver again has a module-simple zero representation. Translation in its trivial factor and another rescaling identify finer limits. The result can be iterated through any fixed finite succession of convergent centers and scales, under the explicit bounded normal-parameter and base-point hypotheses of Theorem 6.4. No comparability of the successive positive scales is imposed. We also obtain relative distances to specified conical strata, the square-root contraction profile of compact cores, and matching to the surrounding free wall metric in the regime $\|\zeta\|/\lambda \to 0$, where λ is the least positive finite orbit eigenvalue.

1.3. **The K3 application.** We realize a projective K3 surface with $\mathrm{NS}(X) = \langle 2\rangle \oplus A_2(-2)$ and an ample class H. Three spherical line bundles with classes $a, b, -a-b$ have four cross-extension directions for every pair. With multiplicities $(2,2,1)$ their direct sum has primitive Mukai vector $(5, a+b, -5)$ of square 46. The resulting local moduli have complex dimension 48. The real polarization family $H + ta + ub$ realizes two independent stability parameters.

The finite representation is a doubled triangle on which two matrix groups $U(2)$ act nontrivially and interact. Five realized wall lines meet at the central bundle and divide the parameter plane into ten chambers. Explicit adjoint-paired arrow matrices prove the existence of simple representations at zero moment level for the full vector and every required wall summand. At selected actual two-factor wall strata, the transverse contractions have models $T^*\mathbb{P}^9$, $T^*\mathbb{P}^{11}$ and $T^*\mathbb{P}^{17}$, with complete product metric limits and precise square-root contraction scales. At the wall the distance to the two-factor stratum is $(1+O(r))\sqrt{2s}$, where s is the nonzero singular value of the rank-one square-zero transverse invariant. Theorem 7.2 states these conclusions for the actual harmonic metric.

The cycle, interacting matrix symmetries, and independent parameters are essential features of this application. There is no vertex-scalar sign acting as $-I$ on every edge, and approaches to its intersecting walls require the singular and iterated limit arguments. The comparison is therefore used beyond a change of matrix size in a two-factor quotient.

1.4. **Relation to the literature and scope.** The quadratic K3 deformation model is classical in origin. Budur–Zhang prove the required formality [BZ19]; Arbarello–Saccà give the equivariant analytic quadratic identification and its quiver description [AS18, Propositions 4.4 and 6.1]. Their dimension-one sheaf wall-crossing theorem is not used to identify this rank-five bundle problem. Their later treatment of Bridgeland moduli compares nearby stability conditions with local quiver VGIT, away from totally semistable walls and for sufficiently close conditions with the same heart [AS25, §6, Claim 6.1, Proposition 6.3 and Corollary 6.4]. This broader local comparison concerns algebraic stability and does not supply the harmonic metric estimates studied here. King's quotient theory [Kin94], Crawley-Boevey's moment-fiber theorem [CB01], and Bellamy–Schedler's normality and projective face maps [BeSc21] supply the algebraic geometry used here. Mayrand's full Hamiltonian slice [May22] supplies the analytic local form; we additionally prescribe its isometric normal first jet and compare its real quotient metric.

Buchdahl–Schumacher's analytic GIT and Weil–Petersson work [BS21, BS23, BS26] provides analytic metric antecedents. Clarke–Tipler prove local HYM wall variation for simple sufficiently smooth semistable bundles with abelian graded automorphism group [CT24, Proposition 1.1 and Theorem 1.2]. Delloque removes the abelian restriction and proves continuity of HYM connection classes under metric variation [Del25a, Theorems 4.8–4.9]. These results concern connections and graded bundles; the relative harmonic moduli-metric estimates require the additional arguments here. We use Delloque's later parameter-dependent local solution theory [Del25b] on its integrable HYM branch for full local classification and inverse coverage, and prove separately the uniform estimates through the retained symmetry directions. Namikawa–Odaka establish metric tangent-cone asymptotics under their reduction hypotheses [NO26, Setup 1, Proposition 6.10 and Corollary 6.13]. Their Remark 6.14 distinguishes the metric tangent cone from the flat symplectic-slice reduction metric, and the derivative estimate in Corollary 6.13(ii) has an isolated-singularity hypothesis. Our conclusions concern the stated quiver and bundle settings, with nonisolated strata and varying real parameters; they are not a general hyperkähler equivalence theorem for arbitrary symplectic singularities. Xu–Zhuang, with an appendix by Guenancia, prove formal conicality and derive analytic conicality in their symplectic-singularity setting [XZ26, Theorem 1.2 and Appendix A]; general hyperkähler-cone existence remains their Conjecture 1.3. These algebraic and analytic conclusions do not by themselves give the quantitative harmonic-metric comparison proved here.

All moduli identifications and contraction maps asserted here are local. In particular we do not identify a global metric completion, a bubbling compactification, or an HYM evolution. The full stratified length metric is specified explicitly; identifying it with the completion of the regular locus alone would be a further statement.

2. Uniform correction in complex quiver orbits

Let Q be a finite quiver, allowing loops and multiple arrows. For Hermitian spaces W_i of dimensions $m_i > 0$, put

$$V = \bigoplus_{a:i\to j} \operatorname{Hom}(W_i, W_j), \qquad K = \prod_i U(W_i), \qquad G = \prod_i GL(W_i).$$

We use the Hilbert–Schmidt metric on V and the ordinary trace pairing on the real vector space $\mathfrak{a} = \bigoplus_i \operatorname{Herm}(W_i)$. The action and its Hermitian infinitesimal action are

$$(gx)_a = g_j x_a g_i^{-1}, \qquad (C_x S)_a = S_j x_a - x_a S_i.$$

Thus the vector field generated by iS is $\xi_S(x) = iC_x S$. Our real moment map and symplectic normalization are

$$\mu_i(x) = \sum_{h(a)=i} x_a x_a^* - \sum_{t(a)=i} x_a^* x_a, \qquad d\langle \mu, S\rangle = 2\omega_0(\xi_S, \cdot), \quad \omega_0(iC_x S, C_x S) = \|C_x S\|^2. \tag{1}$$

The common scalar acts trivially. A central covector is written $\zeta = (\zeta_i I_{W_i})$ with $\sum_i m_i \zeta_i = 0$. A *subrepresentation* is a collection $D_i \subset W_i$ with $x_a D_{t(a)} \subset D_{h(a)}$ for every arrow. A representation is *module-simple* if its only subrepresentations are zero and itself. This condition is independent of a stability parameter and is stronger than having only scalar endomorphisms.

Let Ω be a smooth closed K-invariant two-form on the Euclidean ball $B_R \subset V$, and let $\rho : B_R \to \mathfrak{a}$ be an equivariant moment map with

$$\rho(0) = 0, \qquad d\langle \rho, S\rangle = 2\Omega(\xi_S, \cdot), \qquad \|\Omega - \omega_0\| \le \varepsilon. \tag{2}$$

The last norm is the Euclidean operator norm. We do not assume that Ω is of type $(1,1)$. Its indicated bound, with $\varepsilon < 1$, implies positivity on the pairs (iC_xS, C_xS) used below. We call x ρ-balanced at ζ if $\rho(x) = \zeta$; flat balancing means $\mu(x) = \zeta$.

Theorem 2.1. *There are constants $A, \varepsilon_0 > 0$, depending only on $M = \sum_i m_i$, with the following properties whenever (2) holds and $\varepsilon < \varepsilon_0$.*

1. *If $x \in B_{R/4}$ and $\mu(x) = \zeta$ is central, there is $S \in \mathfrak{a}$ with $\|S\|_{\mathrm{op}} \le A\varepsilon$ and $\rho(e^S x) = \zeta$.*
2. *If $y \in B_{R/4}$ and $\rho(y) = \zeta$ is central, there is $T \in \mathfrak{a}$ with $\|T\|_{\mathrm{op}} \le A\varepsilon$ and $\mu(e^T y) = \zeta$.*
3. *In a fixed complex orbit, the ρ-balanced representatives in B_R at the central value ζ form at most one K-orbit. The corresponding local maps between balanced compact quotients are continuous and inverse on nested neighborhoods. These conclusions remain valid after restriction to a G-invariant analytic subset of V.*

No bound on a positive eigenvalue of $C_x^*C_x$ is required.

The proof extends the spectral-cut and degree argument of [Wue26a, Wue26c]. The projection estimate used in [Wue26d] is the appropriate error bound: both the coercive term and the error contain the same crossing energy. Retaining the full stabilizer in the argument also gives the reverse assertion and continuity when stabilizers change.

2.1. **A relative estimate for every projection.** For a tuple $P = (P_i)$ of orthogonal projections, set

$$q_P(x) = \|C_xP\|^2 = \sum_{a:i\to j} \|P_j x_a - x_a P_i\|^2.$$

Lemma 2.2. *Under (2),*

$$|\langle \rho(x) - \mu(x), P\rangle| \le \varepsilon q_P(x) \qquad (x \in B_R).$$

Proof. Write each arrow in its preserving and crossing blocks for P, and write $x = n + u$ for the resulting orthogonal decomposition. The circle $\exp(itP)$ fixes n. On its fixed vector space both moment components paired with P have zero differential, and their normalizations at the origin are zero. Along $n + su$ we have $\xi_P = s\, iC_uP$ and $\|C_uP\| = \|u\|$. Orthogonality keeps this segment in B_R. The moment identities give

$$\langle \rho(x) - \mu(x), P\rangle = 2\int_0^1 s(\Omega - \omega_0)_{n+su}(iC_uP, u)\, ds.$$

Its absolute value is at most $\varepsilon\|u\|^2 = \varepsilon q_P(x)$. □

2.2. **The stabilizer and a strict spectral cut.** Fix x, let $H = K_x$, and define

$$\mathfrak{h} = \{S \in \mathfrak{a} : C_xS = 0\}, \qquad \mathfrak{c} = \{S \in \mathfrak{a} : Sh = hS \text{ for all } h \in H\}, \qquad \mathfrak{p} = \mathfrak{c} \cap \mathfrak{h}^\perp.$$

The commutation condition is understood vertexwise. If $S \in \mathfrak{p}$, then $e^S x \in V^H$. Any normalized equivariant moment map takes its values on $V^H \cap B_R$ in $\mathfrak{p}$: equivariance puts the value in $\mathfrak{c}$, and the vanishing generating fields of $\mathfrak{h}$ show that its pairing with $\mathfrak{h}$ is constant, hence zero, on this connected fixed ball. In particular a central target attained at x belongs to $\mathfrak{p}$. Order the union of the spectra of all vertex matrices S_i, including multiplicities. Adjacent eigenvalues below refer to this single ordered list.

Lemma 2.3. *If $S \in \mathfrak{p}$ and $\|S\|_{\mathrm{op}} = a > 0$, there is a global adjacent eigenvalue gap of size at least $a/(M-1)$. For the vertexwise spectral projection P above any strict global gap, $q_P(x) > 0$.*

Proof. The identity tuple belongs to $\mathfrak{h}$, so $\operatorname{tr} S = 0$. The oscillation of its M eigenvalues is at least a, proving the gap estimate. A spectral projection P commutes with H. If $q_P(x) = 0$, it also belongs to $\mathfrak{h}$. Since $I - P \in \mathfrak{h}$, orthogonality gives $\operatorname{tr}(SP) = \operatorname{tr}(S(I-P)) = 0$. The mean eigenvalues above and below a strict gap cannot both vanish. □

For $M = 1$ the action is trivial and all normalized moments vanish. The following proof therefore assumes $M \geq 2$.

2.3. **The forward and reverse degree arguments.**

Proof of Theorem 2.1. When $\varepsilon = 0$, the moment identities and the common normalization give $\rho = \mu$, so take zero correction. Assume $\varepsilon > 0$. We may take

$$A = 2M, \qquad \varepsilon_0 = \frac{\log 2}{16A}, \qquad a = A\varepsilon.$$

If $\mathfrak{p} = 0$, both moments at the starting point vanish and no correction is needed. Otherwise consider, on the operator-norm ball of radius a in $\mathfrak{p}$,

$$F_s(S) = \mu(e^S x) - \zeta + s(\rho - \mu)(e^S x), \qquad 0 \leq s \leq 1.$$

This homotopy takes values in the same fixed space $\mathfrak{p}$. All its arguments have norm at most $e^{2a}\|x\| < R/2$.

At a boundary point diagonalize S vertexwise and choose P above a gap $\delta \geq a/(M-1)$. An arrow entry with eigenvalue difference d has squared norm multiplied by e^{2d}. For an entry crossing into P the contribution to the paired moment increment is $(e^{2d} - 1)w$, with $d \geq \delta$; for one crossing out it is $(1 - e^{2d})w$, with $d \leq -\delta$. Therefore

$$\langle \mu(e^S x) - \mu(x), P\rangle \geq (1 - e^{-2\delta}) q_P(x) \geq 2e^{-4a} \frac{a}{M-1} q_P(x), \tag{3}$$

$$|\langle (\rho - \mu)(e^S x), P\rangle| \leq \varepsilon e^{4a} q_P(x). \tag{4}$$

Here $\delta \leq 2a$, and P commutes with S. Our constants make the first coefficient strictly larger than the second. Lemma 2.3 makes the crossing energy positive. Thus F_s has no boundary zero; no continuous choice of P is needed.

The only zero of F_0 is $S = 0$. Indeed

$$\frac{d}{dt}\langle \mu(e^{tS}x), S\rangle = 2\|C_{e^{tS}x} S\|^2$$

is strictly positive for $S \in \mathfrak{p} \setminus \{0\}$: $C_{e^{tS}x} S = e^{tS} C_x S$, and $C_x S = 0$ would imply $S \in \mathfrak{h} \cap \mathfrak{h}^\perp$. Moreover $DF_0(0)$ has positive quadratic form $2\|C_x S\|^2$ on $\mathfrak{p}$, so its local degree is $+1$. Brouwer degree gives a zero of F_1. This uses invertibility at each fixed x, not a uniform lower eigenvalue estimate.

For the reverse assertion set $E = \rho - \mu$, define $\mathfrak{p}$ using $H = K_y$, and use

$$G_s(S) = \mu(e^S y) - \mu(y) - sE(y).$$

All terms again lie in $\mathfrak{p}$. Its initial zero is unique and nondegenerate by the preceding calculation, which does not require $\mu(y)$ to be central. On the boundary (3) dominates $|\langle E(y), P\rangle| \leq \varepsilon q_P(y)$. A zero at $s = 1$ satisfies $\mu(e^S y) = \rho(y) = \zeta$. This proves existence of a flat balanced representative without an independent polystability assumption.

For uniqueness suppose $y, gy \in B_R$ are ρ-balanced at the same central value. Write $g = ke^T$. Equivariance and centrality reduce to $y, e^T y$. In an eigenbasis of T, the squared norm of

$e^{sT}y$ is a sum of nonnegative exponentials, hence convex on $[0,1]$. The entire path therefore stays in B_R, and

$$\frac{d}{ds}\langle \rho(e^{sT}y), T\rangle = 2\Omega(iC_{e^{sT}y}T, C_{e^{sT}y}T) \geq 2(1-\varepsilon)\|C_{e^{sT}y}T\|^2.$$

Its endpoint values agree, so $C_yT = 0$, and $gy = ky$.

Both corrections remain in any invariant analytic subset. Uniqueness makes the compact-orbit maps independent of all choices. For continuity choose convergent representatives and bounded correction matrices. Every convergent subsequence of those matrices supplies a balanced limit in the same complex orbit; uniqueness identifies its compact orbit. The limiting matrix need not lie in the limiting $\mathfrak{p}$. Compactness in the full fixed Hermitian algebra suffices. The same proof gives parameter continuity for smooth families with a common bound.

For explicit domains write $\mathcal{A}_s = (Z \cap \mu^{-1}(\zeta) \cap B_s)/K$ and $\mathcal{B}_s = (Z \cap \rho^{-1}(\zeta) \cap B_s)/K$. The forward and reverse maps send radius $R/4$ into radius $R/2$ and are inverse wherever both are defined. In particular the forward map identifies $\mathcal{A}_{R/8}$ with the open subset $g^{-1}(\mathcal{A}_{R/8})$ of $\mathcal{B}_{R/4}$, where g denotes the reverse map; its image contains $\mathcal{B}_{R/16}$. This proves the local statement with the required slack in radii. □

Corollary 2.4. *For a jointly smooth family satisfying $\|\Omega_t - \omega_0\|_{B_{2r}} \leq C(|t| + r^\alpha)$, the correction on a fixed inner subball has norm $O(|t| + r^\alpha)$, uniformly in stabilizer type and ratios of nonzero arrow sizes. Smooth normalization gives $\alpha = 1$. The value $\alpha = 2$ follows if the first spatial derivative vanishes, for example under a verified inversion symmetry.*

Proof. Apply Theorem 2.1 on B_{2r}. Taylor's theorem gives the two stated rates under their respective derivative hypotheses. □

The theorem alone asserts neither differentiability across strata nor a metric comparison. Both require identifying genuine tangent vectors and removing complex gauge directions; we do this next.

3. Harmonic coordinates and the metric on K3 bundle moduli

Let X be a projective K3 surface and H an integral ample class. Let $F_1, \ldots, F_l$ be pairwise nonisomorphic locally free H-slope-stable sheaves with the same reduced H-Hilbert polynomial. Fix positive multiplicities m_i and put

$$E_0 = \bigoplus_i F_i \otimes W_i, \qquad W_i = \mathbb{C}^{m_i}.$$

For the central analytic background choose a real Kähler class ω_* for which the same F_i are slope-stable of a common slope, and use its Ricci-flat Kähler metric. One may take $\omega_* = H$. These are separate hypotheses: H supplies algebraic formality, whereas ω_* supplies the Hermitian–Einstein background. Choose its direct-sum Hermitian–Einstein metric h_0 on E_0. Fix the determinant, and let ω_t be a smooth finite-dimensional family of Kähler forms with $\omega_0 = \omega_*$.

Give $V = \operatorname{Ext}^1(E_0, E_0)$ its central harmonic Hermitian norm. Serre duality identifies it unitarily with a doubled-quiver representation for $K = \prod_i U(W_i)$; self-extensions give paired loops, as proved below. Let $\mu_{\mathbb{C}}, \mu$ be its complex and real moments with the normalization of (1), and put $Z = \mu_{\mathbb{C}}^{-1}(0)$. For $r_i = \operatorname{rk} F_i$ and $N = \sum_i r_i m_i$, define

$$\zeta_i(t) = 2\pi\big(r_i \mu_{\omega_t}(E_0) - \deg_{\omega_t} F_i\big), \qquad \sum_i m_i \zeta_i(t) = 0. \tag{5}$$

Degrees mean $\int_X c_1(F_i) \wedge \omega_t$ and slope is degree divided by rank. In particular (5) has no volume denominator. This sign agrees with the incoming-minus-outgoing convention: a subbundle of smaller slope has positive total ζ-weight.

At a balanced quiver point, decompose the representation into pairwise nonisomorphic stable summands. Its *representation type* is the unordered list of their dimension vectors and multiplicities. Stability here means strictly positive ζ-weight on every nonzero proper subrepresentation; Proposition 4.1 justifies this decomposition. The connected strata of a fixed type are the orbit-type strata for the compact group action. We use these strata, including singular types, throughout the metric statements.

Theorem 3.1. *Under these hypotheses there are common nested neighborhoods in which the finite quotients*

$$M_{\zeta(t)} = (Z \cap \mu^{-1}(\zeta(t)))/K$$

identify with open neighborhoods in the actual fixed-determinant Hermitian–Yang–Mills bundle moduli. Every sufficiently small actual solution for sufficiently small t is covered. The maps f_t are homeomorphisms, smooth on representation-type strata, and preserve their complex structures. On genuine stratum tangents at a finite representative x, with $r = \|x\|$,

$$(1 - C(|t| + r))g_{\zeta(t)} \le f_t^* g_t^{L^2} \le (1 + C(|t| + r))g_{\zeta(t)}. \tag{6}$$

The constants are independent of stabilizer type and of ratios of positive eigenvalues of the finite orbit operator.

If a central unitary automorphism acts as $-I$ on all of V, the spatial error in (6) improves to Cr^2. At the central value the construction identifies the full reduced analytic quotient germ, including its singular locus.

Here a neighborhood of actual solutions means solutions represented by small deformations of the fixed smooth bundle and central Dolbeault operator, uniformly in t. The central holomorphic bundle itself need not remain polystable when $t \neq 0$. The proof combines a fixed elliptic inverse with Theorem 2.1. The elliptic inverse removes the whole parallel algebra; no potentially small finite symmetry is included in that inverse. The complementary minimization method of [Wue26b, Wue26c, Wue26d] then yields the metric estimate.

3.1. **Unitary quiver coordinates.** Write $d = \bar\partial_0$ on $\operatorname{End} E_0$ and identify V with its harmonic degree-one forms. Fix a parallel holomorphic two-form Ω_X with normalization compatible with the harmonic norm. In unitary coframes and factor frames define

$$J_{ij}(A_1\bar\theta^1 + A_2\bar\theta^2) = -A_2^*\bar\theta^1 + A_1^*\bar\theta^2.$$

This is frame-independent, antiunitary, and maps the (i,j) factor space to the (j,i) space, with $J_{ji}J_{ij} = -I$. Choose the orientation of the Serre pairing $Q(\alpha,\gamma) = \int_X \Omega_X \wedge \operatorname{tr}(\alpha \wedge \gamma)$ so that

$$Q(\alpha, J\beta) = \langle \alpha, \beta \rangle_{L^2}.$$

Pointwise this is the identity $\operatorname{tr}(A_1B_1^* + A_2B_2^*)$. If β is harmonic, its orthogonality to exact forms and Stokes' formula show $dJ\beta = 0$. Apply the opposite pairing, $J^2 = -I$ and Stokes' formula to show that $J\beta$ is orthogonal to exact forms as well. Thus J preserves harmonicity; flat factor connections are not assumed.

Paired orthonormal bases identify off-diagonal extension spaces with opposite arrows. Each diagonal $\operatorname{Ext}^1(F_i, F_i)$ is quaternionic; choose pairs $\alpha, J\alpha$, each giving loop coefficients $A, B \in \operatorname{End} W_i$. The quadratic Yoneda contribution of this pair is $[A, B]$. In the Serre-dual target paired with the parallel algebra by ordinary multiplicity trace, the total quadratic obstruction

is exactly $\mu_{\mathbb{C}}$. If η_X generates $H^{0,2}(\mathcal{O}_X)$ with $\int_X \Omega_X \wedge \eta_X = 1$, its literal harmonic block representative is instead

$$\bigoplus_i r_i^{-1} I_{F_i} \otimes \mu_{\mathbb{C},i}\, \eta_X. \tag{7}$$

The factors r_i^{-1} compensate for tracing I_{F_i}. The ordinary finite dual and these harmonic block coefficients must not be interchanged.

3.2. **Convergent quadratic integrability coordinates.** The harmonic Kuranishi embedding on a full ball is the convergent solution of

$$\Phi(b) = b - d^* G_2(\Phi(b) \wedge \Phi(b)), \qquad D\Phi(0) = \iota. \tag{8}$$

Here G_2 is the Green operator for the degree-two Dolbeault Laplacian at the central bundle. The elliptic contraction construction is holomorphic, and its Green operators commute with the parallel multiplicity action. Its integrability locus has a reduced analytic germ $S \subset V$. Reduced analytic semiuniversality is provided by [BS21, §5 and Remark 4].

The bundle E_0 is H-Gieseker polystable. Budur–Zhang [BZ19, Theorem 1.2] gives formality of its derived endomorphism algebra. The equivariant analytic result of Arbarello–Saccà [AS18, Proposition 4.4], with the quiver identification in their Proposition 6.1, identifies the reduced Quot slice with the quadratic germ Z, tangent to the identity. This is a fixed-bundle germ result. We do not invoke their dimension-one sheaf wall-crossing theorem for vector bundles.

We record the analytic comparison with the harmonic family, because formality alone would not supply controlled harmonic coordinates. Twist the harmonic family by a sufficiently large multiple of H. Analytic base change makes its spaces of sections a vector bundle with surjective evaluation map [Gro61a, Corollaries 1.4–1.5 and Theorem 3.1]. Average a holomorphic frame over K; it remains a frame near the origin and makes the evaluation equivariant. The analytic Quot construction [Gro61b, Theorem 1.1] gives a classifying map. An equivariant complement to the stabilizer in the ambient general linear algebra, followed by Luna's slice [Lun73], gives a K-equivariant germ map $h : S \to Z_{\mathrm{Quot}}$ with identity Kodaira–Spencer derivative.

The reduced tangent space of Z is all of V: every subspace supported on a single arrow, including a single loop with its paired loop zero, lies in Z, and these subspaces span V. First apply reduced semiuniversality to the Quot family to obtain a reverse map $g : Z_{\mathrm{Quot}} \to S$. Its Kodaira–Spencer derivative, followed by harmonic inclusion, is the identity on V. Thus the reduced tangent of S is also V. This argument does not assert that an arbitrary matrix-valued harmonic self-extension squares to zero as a differential form.

Both compositions have invertible tangent maps and are analytic germ isomorphisms. For completeness, lift a self-map with invertible tangent map to an ambient local biholomorphism in a minimal embedding. Its pullback on the germ's local ring is surjective. A surjective endomorphism of a Noetherian ring is injective: the ascending kernels of its powers stabilize, and surjectivity then forces its first kernel to vanish. This proves the assertion, and hence h is an isomorphism.

Extend the resulting coordinate functions to an ambient holomorphic map and average over K. We obtain an equivariant local biholomorphism F satisfying

$$\begin{aligned} &F(x) = x + O(\|x\|^2), \quad DF(x) = I + O(\|x\|), \\ &a(x) = \Phi(F(x)), \qquad\qquad D_x = d + [a(x), \cdot], \qquad J_x = Da(x). \end{aligned} \tag{9}$$

Its integrable locus is exactly Z on suitably small representatives. Compact equivariance complexifies locally, and for every parallel complex endomorphism ξ gives the exact identity

$$D_x\xi = -J_xC_x\xi. \tag{10}$$

For a genuine tangent v to an integrable stratum, $D_x(J_xv) = 0$, by differentiation of integrability.

If formality or the convergent analytic comparison is unavailable, one must retain the full higher-order Kuranishi obstruction. Neither a quadratic tangent obstruction nor a formal equivalence alone justifies the homogeneous quotient and its rescalings.

3.3. **The fixed complementary gauge equation.** Use h_0 and its density dV_0 to define the Hermitian moment

$$\mathcal{M}_t = i(F^0 \wedge \omega_t)/dV_0,$$

where F^0 is the trace-free curvature of the unitary connection determined by a Dolbeault operator. Let Q_0 be the fixed L^2-orthogonal complement of the entire parallel trace-free Hermitian algebra. The derivative in s at the origin of

$$(t, x, s) \longmapsto Q_0\mathcal{M}_t\big(e^s \cdot (\bar\partial_0 + a(x))\big)$$

is a fixed nonzero multiple of the covariant Laplacian on Q_0. All of its kernel has been removed. On fixed Sobolev spaces $W^{k+2,2} \to W^{k,2}$, $k > 2$, the implicit-function theorem gives a unique small smooth $\sigma(t, x) \in Q_0$. Harmonicity and the Kähler identities kill its first x derivative at $(0, 0)$. Elliptic regularity gives, for every fixed j,

$$\|\sigma(t, x)\|_{C^j} \le C_j(|t| + \|x\|^2). \tag{11}$$

Uniqueness gives K-equivariance. Only this fixed elliptic complement is inverted.

Put $\Psi_t(x) = e^{\sigma(t,x)} \cdot (\bar\partial_0 + a(x))$. For $S \in \bigoplus_i \operatorname{Herm} W_i$ use the trace-free lift

$$\widehat{S} = \bigoplus_i I_{F_i} \otimes (S_i - c(S)I_{W_i}), \qquad c(S) = \frac{\sum_i r_i \operatorname{tr} S_i}{N}.$$

Define the finite moment m_t in the ordinary multiplicity dual by

$$\sum_i \operatorname{tr}(m_{t,i}(x)S_i) = \int_X \operatorname{tr}(\mathcal{M}_t(\Psi_t(x))\widehat{S})\, dV_0.$$

Its vanishing is equivalent to vanishing of the remaining parallel component. This is not the rank-and-volume-weighted coefficient of the orthogonal parallel projection. The determinant character on G is $\prod_i(\det g_i)^{r_i}$; using the ineffective common scalar and this lift avoids an incorrect unweighted determinant normalization.

Pullback of the affine unitary-connection symplectic form gives a smooth closed invariant two-form Ω_t on the full ball. In Dolbeault tangent coordinates our affine form is $\operatorname{Im}\langle\alpha, \beta\rangle_{L^2(\omega_t)}$, where the Hermitian pairing is linear in its first argument. Thus the $+iF^0$ convention gives the quadratic moment $+\mu$ and $\Omega_{0,0}(iC_xS, C_xS) = \|C_xS\|^2$. The exact moment identity is $d\langle m_t, S\rangle = 2\Omega_t(\xi_S, \cdot)$. Since $D\Psi_0(0)$ is harmonic inclusion,

$$\Omega_{0,0} = \omega_0, \qquad \|\Omega_{t,x} - \omega_0\| \le C(|t| + \|x\|). \tag{12}$$

This proves positivity on the ordered orbit pairs used in Section 2; compatibility with the original complex structure is not needed.

At $x = 0$, equivariance makes $\sigma(t, 0)$ lie in $\bigoplus_i \operatorname{End} F_i \otimes I_{W_i}$. It need not be scalar on F_i, but preserves the factor decomposition. Chern–Weil therefore gives

$$\langle m_t(0), S\rangle = 2\pi \sum_i \deg_{\omega_t} F_i \operatorname{tr}(S_i - c(S) I_{W_i}) = -\sum_i \zeta_i(t) \operatorname{tr} S_i.$$

The normalized moment $\rho_t = m_t - m_t(0)$ has target $\rho_t = \zeta(t)$. Its complement has already been solved, so this is exactly the remaining projective HYM equation. The sign can also be checked on a single extension arrow $A : F_i \to F_j$. Its moment is negative at i and positive at j. The corresponding extension $0 \to F_j \to E \to F_i \to 0$ can be stable only when $\mu(F_j) < \mu(E)$, precisely the sign $\zeta_j > 0$ in (5). The scalar determinant equation is solved separately. On a K3 surface $H^1(\mathcal{O}_X) = 0$, so fixed determinant removes no scalar degree-one deformation; its Hermitian metric does not alter the endomorphism-valued norm.

Theorem 2.1, applied on a ball proportional to $\|x\|$, now gives forward and reverse corrections of norm $O(|t| + \|x\|)$ in the full multiplicity group. They remain in Z and work for every stabilizer.

3.4. **Complementary minimization and harmonic lengths.** Transport the metric to the uncorrected holomorphic family by $H_{t,x} = h_0 e^{2\sigma(t,x)}$. Let $\mathfrak{g}_0$ be the full parallel complex trace-free algebra and Q its fixed L^2-orthogonal complement in zero-forms. At the central point, $\|\eta\|_{W^{1,2}} \le C\|d\eta\|_{L^2}$ on Q. Smallness of $a(x)$ in C^0 preserves this bound with D_x; nearby Hermitian and Kähler norms are uniformly equivalent. Define by polarization

$$\mathcal{B}_{t,x}(v, v) = \inf_{\eta \in QW^{1,2}} \|J_x v - D_x \eta\|^2_{H_{t,x},\omega_t}. \tag{13}$$

The minimizing normal operator has coercive bilinear form $\langle D_x\eta, D_x\nu\rangle$ on a fixed Hilbert space. Its bounded inverse varies smoothly. Thus $\mathcal{B}$ is smooth, positive near the origin and equivariant, and harmonic orthogonality gives $\mathcal{B}_{0,0} = g_{\text{flat}}$. Consequently

$$(1 - C(|t| + \|x\|))g_{\text{flat}} \le \mathcal{B}_{t,x} \le (1 + C(|t| + \|x\|))g_{\text{flat}}. \tag{14}$$

Split every gauge zero-form into its parallel and complementary parts. Equation (10) gives

$$\inf_{\gamma} \|J_x v - D_x \gamma\|^2 = \inf_{\xi \in \mathfrak{g}_0} \mathcal{B}_{t,x}(v - C_x\xi, v - C_x\xi). \tag{15}$$

At an integrable HYM point and a genuine stratum tangent, Hodge theory identifies the left side with the harmonic Kodaira–Spencer norm. A changing gauge adds an exact D_x term and conjugates isometrically, so its derivative disappears from this minimum. We do not identify arbitrary Zariski tangent vectors at a singular coarse point with metric tangent vectors.

For a tangent lift at a flat balanced point, differentiating the real moment equation gives orthogonality to Hermitian orbit directions. At a central value, Hermitian and unitary orbit directions are orthogonal. Hence the usual quotient norm equals

$$\inf_{\xi \in \mathfrak{g}} \|v - C_x\xi\|^2$$

on these balanced stratum tangents. If the correction is locally $y = g(x)x$, its derivative is $g(x)v$ plus a complex vertical vector, which disappears under this infimum. At the point we may take the bounded $g = e^S$ from Theorem 2.1. Its squared ambient norm distortion is between $e^{-4\|S\|_{\text{op}}}$ and $e^{4\|S\|_{\text{op}}}$, and it maps complex orbit directions onto complex orbit directions. Equations (14) and (15) therefore yield (6) without estimating Dg or inverting a small orbit eigenvalue.

A smooth local correction on each stratum exists as follows. Fix a balanced x_0 with compact stabilizer H and bounded correction $S_0 \in \mathfrak{p}_H$, and set $y_0 = e^{S_0}x_0$. Polar convexity

gives $G_{x_0} = H^{\mathbb{C}}$. Because S_0 commutes with $H^{\mathbb{C}}$, we have $G_{y_0} = H^{\mathbb{C}}$ and $K_{y_0} = H$. On V^H the map

$$(T, u) \longmapsto \rho_t(e^T e^{S_0} u) - \zeta(t), \qquad T \in \mathfrak{p}_H,$$

takes values in that fixed space. Its derivative in T at $(0, x_0)$ has quadratic form $2\Omega_{t,y_0}(iC_{y_0}T, C_{y_0}T) > 0$ for $T \neq 0$. It is invertible at this point. The finite implicit-function theorem gives a smooth local choice, and a compact orbit-type slice and uniqueness identify the choices on overlaps. At x_0 it is the bounded correction already used. The local domains may shrink as orbit eigenvalues vanish; the pointwise relative constant does not. The same argument applies to the inverse. The holomorphic harmonic/Quot map and correction within complex orbits preserve the complex quotient structures on these strata.

3.5. **Full coverage and the analytic quotient.** The slice must cover all nearby actual solutions, not merely construct a family of examples. We use the integrable HYM branch of Delloque [Del25b]: the HYM subsolution condition is automatic in §3.2; Proposition 4.3 gives harmonic local classification, Proposition 4.5 the complementary equation, Corollary 5.2 the nearby morphism identification, and Proposition 5.6 the common-parameter inverse. Lemma 7.3 and Theorem 7.4 give an open-image homeomorphism for the integrable solution quotient. The central solution is integrable; stable factors are simple with vanishing cross-Hom, as required. The possible nonintegrable failure of openness is not used.

To match its gauge normalization with ours, let e^s be the positive part of a small gauge from a harmonic representative to an actual solution. Change the harmonic base by e^h with h parallel. The new positive logarithm is

$$L(s, h) = \tfrac{1}{2} \log(e^{-h} e^{2s} e^{-h}).$$

Its parallel projection has derivative $-I$ in h at zero. The finite implicit-function theorem makes L complementary on a common parameter neighborhood. After unitary gauge, uniqueness identifies it with $\sigma(t, x')$. The reverse finite correction then provides a flat balanced representative. Conversely every integrable zero of the corrected moment is projectively HYM, and the determinant equation makes it HYM. High-Sobolev classification and elliptic regularity give the asserted smooth topology and full inverse coverage. Nested neighborhoods are essential; no same-radius truncation or identification of a whole noncompact quotient with a small actual chart is claimed.

At the central value, the equivariant harmonic-to-Quot analytic germ isomorphism descends through the analytic good quotient. Invariant holomorphic functions pull back to invariant holomorphic functions, so the universal property gives an isomorphism of full reduced complex-space germs, not just their strata. The forward and reverse moment correction identify the closed orbits with actual HYM classes. This also applies with H and ω_* distinct under the stated hypotheses. At a generic nonzero level the entire source chart is stable and smooth, so the stratumwise holomorphic identification already identifies its full complex structure. No global comparison of nearby slope and Gieseker stability is assumed: the latter is used at the central point for its analytic germ.

Finally suppose a central unitary acts by $-I$ on V. All ambient constructions are equivariant. The map F is odd and the coefficient matrices of Ω_t and $\mathcal{B}_{t,x}$ are even in x. Taylor expansion improves both errors to $O(|t| + \|x\|^2)$; the same balancing and minimization proofs give the improved rate. This completes the proof of Theorem 3.1.

4. Metric slices and real stability

We first identify the complex quotients at arbitrary real parameters. This also gives a properness estimate for the whole family, which will be needed for intrinsic metric convergence. We then prove a quantitative slice theorem at an arbitrary singular point.

4.1. **Balancing at arbitrary real parameters.** For a quiver representation x of dimension m, a collection $D = (D_i \subset W_i)$ is a subrepresentation if $x_a D_{t(a)} \subset D_{h(a)}$ for every arrow. Let $\eta \in \mathbb{R}^{Q_0}$ satisfy $\eta \cdot m = 0$. We call x η-semistable if $\eta \cdot \dim D \geq 0$ for every subrepresentation, and η-stable if the inequality is strict for every nonzero proper subrepresentation. It is η-polystable if it is a direct sum of η-stable representations, each of total η-weight zero. These definitions use arbitrary real η.

Proposition 4.1. *An orbit Gx meets $\mu^{-1}(\eta)$ if and only if x is η-polystable. The intersection is one K-orbit. For a balanced point, G_x is the complexification of K_x. A stable representation has only scalar endomorphisms.*

Proof. For stable x, consider the left K-invariant function

$$\mathcal{F}(g) = \|gx\|^2 - 2\sum_i \eta_i \log|\det g_i|.$$

Multiplying by a common positive scalar does not change it. Polar decomposition therefore reduces minimization to

$$F(S) = \|e^S x\|^2 - 2\langle \eta, S\rangle, \qquad S \in \mathfrak{a}, \quad \operatorname{tr} S = 0.$$

This function is proper. Indeed, let $\|S_n\| = r_n \to \infty$ and pass to a subsequence with $S_n/r_n \to T$, $\|T\| = 1$. Diagonalize vertexwise and pass to convergent unitary eigenbases and normalized eigenvalues. If a nonzero coefficient of x in the limiting bases goes from a T-eigenvalue τ to a larger τ', its contribution to $\|e^{S_n}x\|^2$ grows at least exponentially in r_n, whereas the trace term is linear.

Otherwise every cumulative lower eigenspace of T is invariant. For its distinct global eigenvalues $\tau_1 < \cdots < \tau_l$, let $D_j = W_{\leq \tau_j}$ and let P_j be the corresponding projection. There are at least two distinct eigenvalues, since T is nonzero and trace-free. Thus

$$T = \tau_l I - \sum_{j<l}(\tau_{j+1} - \tau_j)P_j, \qquad \langle \eta, T\rangle = -\sum_{j<l}(\tau_{j+1} - \tau_j)\eta \cdot \dim D_j < 0.$$

It follows that $F(S_n) \geq -2r_n\langle \eta, S_n/r_n\rangle \to \infty$. These subsequence alternatives prove properness. A minimum exists, and varying $\mathcal{F}$ by $g \mapsto e^{tA}g$ gives $2\langle \mu(gx) - \eta, A\rangle = 0$ for every Hermitian A. The scalar equation vanishes identically, so $\mu(gx) = \eta$.

For a balanced representation and an invariant D, compression of the moment equations by the orthogonal projection P_D gives

$$\eta \cdot \dim D = \sum_a \|P_{D_{h(a)}} x_a (1 - P_{D_{t(a)}})\|^2 \geq 0. \tag{16}$$

Equality means that D is reducing. Split orthogonally whenever equality occurs for a proper nonzero D and induct on total dimension. This gives stable summands of zero total weight. Conversely balance each stable summand by the preceding minimization and take their orthogonal direct sum.

Uniqueness follows from polar decomposition and convexity as in Theorem 2.1: for balanced $x, e^S x$ the function $\|e^{tS}x\|^2 - 2t\langle \eta, S\rangle$ has zero endpoint derivatives and second derivative

$4\|C_{e^{tS}x}S\|^2$, so $e^S x = x$. If g stabilizes x, its compact and positive polar factors therefore both stabilize x, proving $G_x = (K_x)^{\mathbb{C}}$.

Finally, a nonzero singular endomorphism of a stable representation would have nonzero proper kernel and image, both of positive weight, while their weights add to zero. Every nonzero endomorphism is therefore invertible. Subtracting an eigenvalue proves it scalar. □

The proposition is the polystable form of the real Kempf–Ness criterion. Its direct proof avoids any genericity restriction. In particular, a balanced orbit with a positive-dimensional stabilizer is not incorrectly declared stable.

4.2. **Invariant properness and fixed faces.** From now on V is a doubled-quiver representation, with $Z = \mu_{\mathbb{C}}^{-1}(0)$ given its reduced structure. Choose positive-degree homogeneous generators $f_1, \dots, f_q$ of $\mathbb{C}[Z]^G$, of degrees d_j, and set $a(x) = \max_j |f_j(x)|^{1/d_j}$; set $a = 0$ if there are no generators. Finite generation is the standard reductive invariant-theory theorem [KP96, §1.6].

Lemma 4.2. *There is a constant C such that*

$$\|x\| \le C\big(\|\mu(x)\|^{1/2} + a(x)\big), \qquad x \in Z. \tag{17}$$

Consequently the family of balanced compact quotients maps properly to the product of the central real parameter space and $Z/\!/G$. Representatives in the affine fiber over the origin satisfy $\|x\| \le C\sqrt{\|\eta\|}$.

Proof. The right-hand parenthesis is strictly positive on the unit sphere in Z. A zero there would be balanced at zero and have all invariant coordinates equal to those of the origin. Zero-character Kempf–Ness makes its complex orbit closed, and the affine quotient fiber has the origin as its unique closed orbit [Kin94, §2 and Theorem 6.1], a contradiction. Its positive minimum and real homogeneity prove the estimate. Bounded parameter and invariant coordinates then bound all balanced representatives. Their inverse image of a compact set is closed before quotienting, hence compact. The last assertion follows when all f_j vanish. □

Partition $\Theta = \{\eta : \eta \cdot m = 0\}$ by the signs of $\eta \cdot d$ for every integral $0 \le d \le m$, excluding $0, m$. Each relatively open face F is rational polyhedral and contains a rational parameter; clear denominators to choose an integral $\theta_F \in F$. Our character is $\chi_{\theta_F}(g) = \prod_i \det(g_i)^{(\theta_F)_i}$. This is King's positive subrepresentation convention [Kin94, Propositions 3.1 and 3.2]. All stable, semistable and polystable objects are constant on F: the sign list includes dimensions of every possible summand and its subrepresentations.

Corollary 4.3. *For $\eta \in F$, the fixed algebraic quotient map induces a homeomorphism*

$$M_\eta = (Z \cap \mu^{-1}(\eta))/K \;\longrightarrow\; Q_F = Z/\!/_{\chi_{\theta_F}} G.$$

Proof. King's description of the closed semistable orbits and Proposition 4.1 give a continuous bijection. Over a compact subset of Q_F, affine invariant coordinates are bounded, so Lemma 4.2 bounds representatives. Every convergent sequence of them has an η-balanced limit, which is still θ_F-polystable. The quotient map is therefore defined at the limit and takes it into the same compact set. The bijection is proper, hence a homeomorphism. □

We shall use the fixed algebraic quotient maps on representative spaces. The corollary does not by itself give a jointly continuous inverse as the real parameter varies.

4.3. **A quantitative Hamiltonian slice.** Fix x_* with $\mu_{\mathbb{C}}(x_*) = 0$ and $\mu(x_*) = \eta_*$ central, and let $H = K_{x_*}$. The quaternionic normal representation is

$$W = (L \oplus JL)^{\perp}, \qquad L = T_{x_*}(Gx_*),$$

where J is the antiunitary quaternionic structure on V. The complex orbit tangent L is holomorphically symplectic isotropic, $L \perp JL$, and $L \oplus JL$ is quaternionic. Thus W inherits its Hermitian and holomorphic symplectic forms.

Theorem 4.4. *There are nested common neighborhoods, for η near η_*, identifying M_η near $[x_*]$ with the quotient of the full normal representation W at central value $\eta|_{\mathfrak{h}}$. Here $\mathfrak{h}$ denotes the Hermitian stabilizer and restriction is dual to its actual embedding in the vertex algebra. The identifications are homeomorphisms onto open neighborhoods, smooth on strata, and their quotient metrics satisfy*

$$(1 - C(|\eta - \eta_*| + \|w\|))g^W_{\eta|_{\mathfrak{h}}} \le f_\eta^* g_\eta \le (1 + C(|\eta - \eta_*| + \|w\|))g^W_{\eta|_{\mathfrak{h}}}. \tag{18}$$

Constants depend on the fixed center x_*, but not on positive eigenvalues of the retained H-orbit operator at w. The representation W is unitarily a doubled Ext-quiver representation, including paired loops.

Proof. Shift the real moment by η_* and apply the full holomorphic Hamiltonian slice theorem of Mayrand [May22, Theorem 1.3, Proposition 3.1 and (3.5)]. The ambient model is

$$G \times_{H^{\mathbb{C}}} (\mathfrak{h}^\circ \times W), \qquad [g, \xi, w] \longmapsto \mathrm{Ad}^*_g(\xi + \mu_{\mathbb{C},W}(w)).$$

The two target summands are complementary, so the complex zero locus is exactly $\xi = 0$, $\mu_{\mathbb{C},W}(w) = 0$. The theorem applies in the shifted semistable open neighborhood; closedness of Gx_* in the original affine space is unnecessary.

We need the slice differential to be the isometric normal inclusion. Identify the orbit directions with L, their symplectic cotangent dual with JL by the fixed orbit Gram pairing, and the normal with the actual W by the identity. This is an H-equivariant symplectic linear identification of full tangents, isometric on W. It can be prescribed in the holomorphic slice: the derivative of an initial slice differs by $(a, n) \mapsto (a + An, Bn)$, and the local map $[g, n] \mapsto [g \exp(An), Bn]$ realizes this change [May22, Theorem 3.9 and §3.5]. The pulled-back and model symplectic forms now agree as full tensors along the zero orbit. Relative radial homotopy produces a primitive of their difference vanishing to second order in normal variables; the equivariant Moser vector field has the same order. Its flow has identity first jet on the zero section and preserves the prescribed normal isometry. This is the relative Moser construction of [May22, Theorem 3.3 and Remark 3.4], with the order of vanishing made explicit. Restricting to $[1, 0, w]$ gives a full-ball holomorphic map F with $F(0) = x_*$ and $DF(0)$ the isometric inclusion.

For the representation claim, decompose the finite-dimensional star algebra generated by vertex projections, arrows of x_* and their adjoints:

$$E_i = \bigoplus_a D_i^a \otimes N_a, \qquad (x_*)_e = \bigoplus_a (x^a)_e \otimes I_{N_a}, \qquad H = \prod_a U(N_a).$$

Proposition 4.1 identifies its complexification with the full complex stabilizer. The factors are stable objects at η_*, not necessarily simple modules. The moment deformation complex splits into factor complexes tensored with $\mathrm{Hom}(N_a, N_b)$. Its orthogonal first cohomology is

$$W = \bigoplus_{a,b} \mathcal{E}_{ab} \otimes \mathrm{Hom}(N_a, N_b),$$

with product Hermitian norm. Quaternionic duality pairs $\mathcal{E}_{ab}$ with $\mathcal{E}_{ba}$ antiunitarily and makes each $\mathcal{E}_{aa}$ quaternionic. Paired orthonormal bases give the unitary doubled Ext quiver, including pairs of loops. This is the metric realization of the usual Ext-quiver slice [BeSc21, §3.2]. If $d_i^a = \dim D_i^a$, restriction of a central covector gives precisely

$$\zeta_a = \sum_i d_i^a \eta_i. \tag{19}$$

Let $\mathfrak{q} = \mathfrak{h}^\perp$ be the fixed Hermitian complement. Solve

$$P_{\mathfrak{q}}\big(\mu(e^T F(w)) - \eta\big) = 0, \qquad T \in \mathfrak{q}.$$

Its derivative at $(\eta_*, 0, 0)$ has positive quadratic form $2\|C_{x_*}T\|^2$, invertible on $\mathfrak{q}$. The implicit-function theorem gives a unique small equivariant $\sigma(\eta, w)$ with $\sigma = O(|\eta - \eta_*| + \|w\|^2)$; the linear w term vanishes by normal orthogonality. Put $\Psi_\eta = e^\sigma F$. At $w = 0$ this is H-fixed. The original homogeneous moment paired with $\mathfrak{h}$ vanishes on the entire H-fixed subspace. Also $\eta_*|_{\mathfrak{h}} = 0$, since $\langle \mu(x_*), S\rangle = \mathrm{Re}\langle C_{x_*}S, x_*\rangle$. Thus the remaining moment $\rho_\eta = \mu(\Psi_\eta)|_{\mathfrak{h}}$ is normalized at zero, with target $\eta|_{\mathfrak{h}} = (\eta - \eta_*)|_{\mathfrak{h}}$.

The form $\Omega_\eta = \Psi_\eta^* \omega_0$ is closed and invariant, has this exact moment map, and differs from the flat form on W by $O(|\eta - \eta_*| + \|w\|)$. Apply Theorem 2.1 to the full retained H quiver. This gives both directions of the finite correction uniformly in its weak orbit eigenvalues.

For the metric, transport the ambient Hermitian norm through e^σ and define on the full tangent space to W

$$B_{\eta,w}(v, v) = \inf_{T \in \mathfrak{q}^{\mathbb{C}}} \|DF(w)v - C_{F(w)}T\|^2_{e^\sigma}.$$

The complementary Gram matrix is invertible at the center and nearby, so B is smooth, equals the flat normal norm at the center, and has error $O(|\eta - \eta_*| + \|w\|)$. Exact equivariance of F and splitting of the full orbit algebra give

$$\inf_{\gamma \in \mathfrak{g}} \|DF(w)v - C_{F(w)}\gamma\|^2_{e^\sigma} = \inf_{\xi \in \mathfrak{h}^{\mathbb{C}}} B_{\eta,w}(v - C_w\xi, v - C_w\xi).$$

On genuine integrability-stratum tangents the left side is the quotient norm. Derivatives of σ and of the retained complex correction are orbit directions and disappear in this infimum. The correction has norm $O(|\eta - \eta_*| + \|w\|)$, so its action distorts squared norms by $1 + O(|\eta - \eta_*| + \|w\|)$. This proves (18).

For clarity, smoothness on a fixed stratum follows from a separate implicit-function argument. At a balanced point with compact stabilizer J, the correction commutes with J, and its complex stabilizer remains $J^{\mathbb{C}}$. On the fixed subspace W^J the remaining moment takes values in the commutant of J orthogonal to its Hermitian algebra. Its derivative in that positive-gauge space has quadratic form $2\Omega(iC_yT, C_yT) > 0$ for $T \neq 0$. The implicit-function theorem gives a smooth local correction there. A compact orbit-type slice and uniqueness identify its local maps on overlaps. Only the inverse at the fixed point is used.

Finally the full complex Hamiltonian slice classifies every nearby integrable orbit. Write its small complex gauge in polar form. Its positive logarithm can be normalized to lie in $\mathfrak{q}$ by varying the positive H factor: if the original logarithm is s, the derivative at zero of the $\mathfrak{h}$ projection of

$$\frac{1}{2}\log(e^{-h}e^{2s}e^{-h})$$

with respect to $h \in \mathfrak{h}$ is $-I$. The finite implicit-function theorem supplies this normalization. Uniqueness of the complementary solution identifies it with σ; the reverse part of Theorem 2.1 then gives a flat normal-balanced representative. All constructions hold on common nested

parameter neighborhoods. Compact-quotient uniqueness proves continuity of the inverse, giving full local coverage. □

Write $U = W^H$ and $W = U \oplus W'$ orthogonally. These are symplectic summands; the moments ignore U. The normal quotient is therefore the exact metric product

$$W/\!/\!/_\zeta H = U \times (W'/\!/\!/_\zeta H). \tag{20}$$

The stratum through the center is U times the normal vertex. In particular, holding a balanced $z \in W'$ fixed while following a path in U has exactly its Euclidean U-length in the model. The full W is a doubled-quiver representation; W' alone may contain traceless loop spaces and need not be one.

5. Intrinsic metrics and varying parameters

Let V be a finite doubled-quiver representation and suppose $Z = \mu_{\mathbb{C}}^{-1}(0)$ contains a module-simple representation of dimension m. Loops and divisible dimension vectors are allowed. Set $A_\eta = Z \cap \mu^{-1}(\eta)$ and $M_\eta = A_\eta/K$. An *admissible path* in M_η is continuous on a compact interval with finite subdivision, smooth on each open subinterval in a representation-type stratum, and of finite one-sided stratum length at every endpoint. Let d_η be the infimum of these lengths. We include all singular strata.

Theorem 5.1. *Every (M_η, d_η) is a nonempty proper geodesic space, with its compact-quotient topology. If $\eta_n \to \eta_*$ and balanced representatives satisfy $x_n \to x_*$, $y_n \to y_*$, then*

$$d_{\eta_n}([x_n],[y_n]) \longrightarrow d_{\eta_*}([x_*],[y_*]). \tag{21}$$

Every target point is a limit of balanced representatives at η_n, after unitary adjustment. For convergent balanced base representatives $p_n \to p_*$, the complete pointed spaces $(M_{\eta_n}, d_{\eta_n}, [p_n])$ converge to $(M_{\eta_*}, d_{\eta_*}, [p_*])$ in pointed Gromov–Hausdorff distance.

The upper distance bound needs two separate facts: uniformly short paths near a singular point and almost isometric paths along strata.

5.1. Semialgebraic paths and inner distances.

Lemma 5.2. *For a fixed semialgebraic family $\mathcal{A} \subset P \times \mathbb{R}^N$ of nonempty compact connected fibers, there is C such that any two points in $\mathcal{A}_p$ have a continuous semialgebraic joining path of length at most $C \operatorname{diam}_{\mathrm{Eucl}} \mathcal{A}_p$.*

Proof. Semialgebraic triviality partitions the base into finitely many pieces with fixed compact model fibers. Triangulate each model. By a finite ordered rule choose simplices containing the endpoints, join each endpoint to a vertex, and join the vertices along a spanning tree of the connected one-skeleton. Transport these paths through the trivializations. The resulting function of parameter, endpoints and time is semialgebraic and continuous in time; joint endpoint continuity is unnecessary. Uniform monotonicity bounds by an integer J the number of monotonicity intervals of every coordinate of every path. Thus

$$\operatorname{Length}(\gamma) \le \sum_{\ell=1}^{N} \operatorname{Var}(\gamma_\ell) \le NJ \operatorname{diam}_{\mathrm{Eucl}} \mathcal{A}_p.$$

The finiteness, triangulation and triviality statements used here are recalled in [vD99, pp. 116, 130, 133, 135]. □

For a closed connected semialgebraic set A, its full Euclidean inner distance δ_A is finite, since semialgebraic arcs join its points. It induces the Euclidean topology locally. At $x \in A$, the component C of $A \cap \overline{B}_R(x)$ containing x is compact and connected and contains a relative neighborhood of x: the finitely many other compact components have positive distance from x. Kurdyka–Orro's inner-metric inequality gives

$$\delta_C(x,y) \le C_0 \|x-y\|^\beta, \qquad 0 < \beta \le 1$$

on C [KO97, Corollary 2]. Since $\|x-y\| \le \delta_A(x,y) \le \delta_C(x,y)$, the assertion follows. Closed δ_A-balls are Euclidean bounded and compact: every Euclidean convergent subsequence converges also in δ_A.

We also require finite-stratum approximations. Apply the near-straight decomposition of [KO97, Proposition 3, pp. 176–178] separately to the finitely many prescribed semialgebraic pieces in a bounded truncation of A. For each $\varepsilon > 0$ it yields finitely many analytic pieces Γ_a, each in one prescribed piece, whose closures F_a cover the truncation and admit C^1 joining arcs with interior in Γ_a and length at most $(1+\varepsilon)$ times Euclidean endpoint distance.

For a rectifiable path, start at its first time, choose a member of the finite closed cover containing that point and visited later, and jump to its last visit. Repeat until the endpoint. Finiteness and closedness ensure a choice exists, and no member can recur after its last visit. The resulting finite chain has total chord length no greater than the original length. Replace each chord by its near-straight arc. Smooth each C^1 arc inside Γ_a on its open interval, using tolerances tending to zero at its endpoints and arbitrarily small integrated derivative error. This preserves endpoints and adds arbitrarily little length. Finite smooth-stratum paths therefore approximate the full inner distance, even for an original path with infinitely many crossings. We do not require geodesics themselves to have finite subdivisions.

5.2. **Fixed singular levels.** The existence of a module-simple representation makes Z reduced, irreducible and generically module-simple [CB01, Theorem 1.2]. A module-simple representation is stable for every parameter, so Proposition 4.1 gives nonemptiness. The complex quotient Q_F is irreducible and normal [BeSc21, Proposition 3.21]; hence M_η is connected by Corollary 4.3. The compact quotient $A_\eta \to M_\eta$ is proper and has connected fibers, since K is connected. Thus A_η is connected: a separation of its domain projects to a separation of the base, because each connected fiber lies in one part and the quotient is closed. Properness here also follows by bounding the descended Euclidean norm over a compact subset of the quotient.

Let δ_η denote the full Euclidean inner metric on A_η. The preceding local argument makes it proper and topologically compatible.

Lemma 5.3. *For $x, y \in A_\eta$,*

$$d_\eta([x],[y]) = \min_{k \in K} \delta_\eta(x, ky). \tag{22}$$

The metric d_η is proper, complete and geodesic.

Proof. Smooth prequotient representation-type pieces map by Riemannian submersions to quotient strata [May22, Theorem 1.1(i), §2.5]. Finite-stratum approximation and projection give the upper bound for d_η in (22).

Conversely lift each open smooth segment of an admissible quotient path horizontally, initializing at an interior time. The lift extends over every compact subinterval: the quotient image is compact in one smooth stratum and its preimage is compact, since K is compact and radius is bounded. The lifted speed equals the quotient speed. Finite endpoint lengths

give Euclidean Cauchy limits in the closed moment level, lying in the prescribed endpoint orbits. Constant unitary adjustments align successive segments and the first initial endpoint with x. The concatenation ends at some ky and has exactly the quotient length, proving the reverse inequality.

The right side is the metric quotient of a proper inner space by a compact isometric group. It is proper and has the compact-quotient topology. The subarc inequality and admissible near-minimizers show it is a length metric. Proper length-space Hopf–Rinow gives geodesicity and completeness. □

In particular $r([x]) = \|x\|$ is 1-Lipschitz. At zero level radial paths are horizontal, so

$$d_0([x], 0) = \|x\|. \tag{23}$$

This concerns the full quotient, not the completion of its regular locus alone.

5.3. **Connected neighborhoods in nearby levels.** Pass to a subsequence on which every η_n belongs to one face F of the finite sign arrangement in Section 4. The face F_* containing η_* lies in its closure. The semistable inclusion gives a projective morphism

$$\pi : Q_F \longrightarrow Q_{F_*} \tag{24}$$

by [BeSc21, Lemma 2.4], with the sign of its character reversed to match our convention. The common dense module-simple locus is stable for every character. Its common geometric quotient is a dense open on which π is an isomorphism. Thus π is birational and surjective by properness. Normality of the integral target gives connected fibers [Sta, Lemma 37.53.6, Tag 0AY8]. These deductions use the existence of a module-simple point in the complex zero fiber.

Fix $p_* = [x_*]$ and its image $q_* \in Q_{F_*}$. In a closed algebraic embedding of an affine open chart containing q_*, let U_s be the component of the closed radius-s ball intersection containing q_*. These are compact connected semialgebraic sets, nested as s decreases, with intersection $\{q_*\}$. Each contains a relative open neighborhood of q_*: the other finitely many compact components are separated from it. Their family is semialgebraic by triviality and uniform finiteness of connected components.

For the fixed algebraic quotient map q_{F_*} on its semistable open set, put

$$B_{\eta,s} = \{x \in A_\eta : q_{F_*}(x) \in U_s\}, \qquad \eta \in F. \tag{25}$$

The domain contains A_η, and this is a semialgebraic family in the original representative space. No inverse balancing map is assumed semialgebraic or jointly continuous. Its compact quotient is $\pi^{-1}(U_s)$, which is compact and connected: a proper surjection with connected fibers has connected inverse image of a compact connected set, by the closed-map separation argument. Applying the same argument to connected K-orbits makes $B_{\eta,s}$ compact and connected.

Furthermore

$$\sup_{x \in B_{\eta,s}} \operatorname{dist}_{\mathrm{Eucl}}(x, Kx_*) \longrightarrow 0 \quad \text{as } (\eta, s) \to (\eta_*, 0), \quad \eta \in F. \tag{26}$$

Estimate (17) bounds the representatives, since their affine invariants range in the compact image of one U_{s_0}. Any contrary sequence has a convergent subsequence with limit $x \in A_{\eta_*}$. This point is F_*-polystable, so the quotient map is defined near it and gives $q_{F_*}(x) = q_*$. Balanced-orbit uniqueness forces $x \in Kx_*$, a contradiction.

Taking representatives over any point of $\pi^{-1}(q_*)$ also proves coverage of the target: they are bounded at η_n and every limiting orbit is Kx_*. Unitary adjustment yields representatives converging to x_*.

Lemma 5.4. *As $(\eta, s) \to (\eta_*, 0)$ with $\eta \in F$, $\operatorname{diam}_{d_\eta}(B_{\eta,s}/K) \to 0$.*

Proof. The stabilizer $H = K_{x_*}$ is a connected product of unitary groups. A small Euclidean normal tube to Kx_* is $K \times_H N_\varepsilon$, where $N = (T_{x_*}Kx_*)^\perp$, under $[k, n] \mapsto k(x_* + n)$. This tube is semialgebraic: its group, orbit, normal space and action are semialgebraic, and the nearest-point inverse is given by polynomial quantifiers. For nearby parameters (26) puts $B_{\eta,s}$ in the half-sized tube. The slice $S_{\eta,s} = B_{\eta,s} \cap (x_* + N_\varepsilon)$ is compact and semialgebraic with $B_{\eta,s} = K \times_H S_{\eta,s}$. It is connected: connected H preserves each of its finitely many open-and-closed components, and multiple components would separate the associated bundle. Its Euclidean diameter tends to zero by the nearest-point property and (26). Lemma 5.2 gives joining paths in this fixed family with length tending uniformly to zero. Projection and (22) give the assertion. □

No connectedness of arbitrary moving moment-level ball intersections has been assumed.

5.4. **Sharp distance bounds and complete pointed limits.**

Proof of Theorem 5.1. For the lower bound choose a subsequence with finite liminf of the distances, and admissible near-minimizers. Lift them from x_n to $k_n y_n$ as in Lemma 5.3. Constant-speed parametrization on $[0, 1]$ gives bounded Lipschitz constants and radius. After subsequences $k_n \to k$ and Arzelà–Ascoli gives a uniform limiting curve in A_{η_*} from x_* to ky_*. Lower semicontinuity of length gives

$$d_{\eta_*}([x_*], [y_*]) \le \liminf_n d_{\eta_n}([x_n], [y_n]).$$

For the upper bound take an admissible limit path of length at most $d_{\eta_*}([x_*], [y_*]) + \varepsilon$. At an interior point of any stratum piece, Theorem 4.4 identifies the stratum with $U \times \{0\}$ in (20). Choose the chart radius so that the squared-metric error is less than ε at η_*. Coverage, followed by the common slice inverse, supplies nearby normal balanced points $(u_n, z_n) \to 0$. Since the moment ignores U, $(0, z_n)$ is balanced.

For a compact limit segment $(u(t), 0)$ use $(u(t), z_n)$ at η_n and apply the slice map. Its model length is the Euclidean length of u, so comparison on both levels gives, for large n,

$$\operatorname{Length}(\gamma_n) \le \sqrt{\frac{1 + 2\varepsilon}{1 - 2\varepsilon}}\, \operatorname{Length}(\gamma), \qquad \varepsilon < 1/4.$$

Its stabilizer is independent of $u(t)$, so it lies in one stratum. Endpoints converge by parameter continuity of the slice and complementary correction: the bounded retained correction acts trivially on U as $z_n \to 0$. No weak-eigenvalue inverse or derivative estimate is used.

Around each original junction choose U_s with nearby intrinsic diameter as small as prescribed by Lemma 5.4. Truncate the incident open pieces so their remaining endpoints are in its relative interior. Continuity and the neighborhood property permit this; finite endpoint lengths make the removed total length arbitrarily small. Cover the remaining compact smooth pieces by finitely many small slice charts and subdivide. Use the sharp length bound on each segment.

At new subdivision points the two nearby endpoints converge to the same target point, so their quotient images lie in some U_{s_n} with $s_n \to 0$ by a diagonal choice of nested neighborhoods. The diameter lemma joins them at cost tending to zero. At original junctions the truncated endpoints and nearby approximations lie in the previously chosen U_s, giving

the prescribed small cost. The same construction joins to $[x_n]$ and $[y_n]$. Constant or wholly truncated pieces use only these joins; a constant whole limit path follows from the diameter lemma. There are finitely many joins, so assign total cost at most ε. Letting $\varepsilon \to 0$ proves the upper bound. Finite faces and the subsequence principle prove (21) for the full sequence.

A bounded ball about $[p_n]$ has representatives in a fixed Euclidean ball because radius is 1-Lipschitz. Every sequence in these balls has a subsequence of convergent representatives, with all limiting distances determined by (21). Conversely coverage supplies nearby representatives for every target point. Choose a finite ε-net in the compact target ball of radius $L+1$. Its nearby representatives form an asymptotic 2ε-net of the varying radius-L balls: an uncovered sequence would have a convergent representative and contradict distance continuity. Finite distance distortions tend to zero. Ball inclusions hold with radii $L \pm o(1)$; boundary points can be approached from inside using geodesics on either side. This proves pointed convergence on every bounded ball to the whole proper geodesic target. □

6. Rescaling, contraction and matching

Let f_t be the local actual-moduli comparison of Theorem 3.1. Write $g_t^{L^2}$ for the actual harmonic metric and $g_{\zeta(t)}$ for the finite quotient metric. We use the estimate

$$(1 - C(|t| + r^\alpha))g_{\zeta(t)} \le f_t^* g_t^{L^2} \le (1 + C(|t| + r^\alpha))g_{\zeta(t)}, \qquad r = \|x\|, \tag{27}$$

where $\alpha = 1$ in general and $\alpha = 2$ under that theorem's verified parity hypothesis. Actual distances use the corresponding finite-stratum curve class in the full bundle-moduli space. Global completeness of that space is not assumed.

6.1. **Localization and distances to wall strata.** Shrink the common chart so that it contains $r < 2R$ and has coarse squared-metric bounds $1/2$ and 2. Each closed radial sublevel is compact in the quotient topology, so its image is compact and closed in the Hausdorff actual moduli space. A path starting at radius $\rho < R$ and escaping the image of $r < R$ must first reach radius R. Its initial segment has length at least $(R-\rho)/\sqrt{2}$, by the 1-Lipschitz radius. While a path of length L stays in the chart, its lifted radius is at most $\rho + \sqrt{2}L$. These observations exclude short paths that leave the local model.

Proposition 6.1. *At $t = 0$, let $\Sigma \subset M_0$ be a closed conical union of representation-type strata containing the vertex. Suppose the specified actual target has intersection with the chart equal to $f_0(\Sigma)$. Then, as $r(x) \to 0$,*

$$d_0^{L^2}(f_0(x), f_0(\Sigma)) = (1 + O(r(x)^\alpha))d_0(x, \Sigma).$$

For the vertex this is $d_0^{L^2}(f_0(x), [E_0]) = r(x)(1 + O(r(x)^\alpha))$.

Proof. The finite distance to Σ is at most $r(x)$, using the vertex. An admissible near-minimizer of length at most $d_0(x,\Sigma) + \delta$ stays within radius $2r(x) + \delta$. Its actual image is a competitor. Any shorter actual competitor has length $O(r(x))$, cannot escape the larger chart, and stays within radius $C_0 r(x)$ by the preceding barrier. Its endpoint cannot lie in a portion of the target outside the chart without paying that exit cost. Apply (27) on these paths and let $\delta \downarrow 0$. The estimate is multiplicative even when the target distance is much smaller than $r(x)$. The vertex formula uses (23). □

The target identification is a geometric hypothesis: a tensor estimate alone does not identify a named singular stratum.

6.2. **Arbitrary directions and complete actual limits.** Assume the finite complex zero fiber contains a module-simple point, as in Section 5.

Theorem 6.2. *Let $t_n \to 0$ and $\delta_n \downarrow 0$, with $\eta_n = \zeta(t_n)/\delta_n^2 \to \eta_*$. Let balanced base representatives $p_n \in A_{\eta_n}$ converge to p_*. With base point $f_{t_n}([\delta_n p_n])$ and distances divided by δ_n, the actual moduli have complete pointed local limit $(M_{\eta_*}, d_{\eta_*}, [p_*])$. On each bounded rescaled ball, the intrinsic comparison with M_{η_n} is relative with error $O_L(|t_n| + \delta_n^\alpha)$.*

Proof. Homogeneity identifies $M_{\delta_n^2\eta_n}$ with $\delta_n M_{\eta_n}$, multiplying distances by δ_n. The fixed chart has an exit cost bounded below independently of n from starting radius $O(\delta_n)$. Any actual path of length $O(\delta_n)$ stays inside it, and its finite radius is $O(\delta_n)$ by the coarse comparison. Conversely, admissible finite near-minimizers between points in a bounded model ball have uniformly bounded radial coordinate by the 1-Lipschitz radius and their length bound. They scale into the actual chart. On both types of paths (27) has the claimed error. Taking infima gives both intrinsic inequalities, including pairs at distances arbitrarily smaller than δ_n.

Full inverse coverage puts every actual radius-$\delta_n L$ ball inside the image of a model radius-$L + o(1)$ ball, and gives the reverse inclusion with radius $L - o(1)$. The model balls are compact; local distance comparison makes these actual balls compact as well. Theorem 5.1 now proves convergence to the entire complete target. No global properness of the actual bundle-moduli space is used. □

Along a fixed ray $\zeta(t_\kappa) = \kappa\eta$ this gives the scale $\sqrt{\kappa}$ and error $O_L(|t_\kappa| + \kappa^{\alpha/2})$. For any specified compact subset $C \subset M_\eta$ of positive diameter,

$$\operatorname{diam}_{L^2} f_{t_\kappa}(\sqrt{\kappa}C) = \sqrt{\kappa}\big(1 + O(|t_\kappa| + \kappa^{\alpha/2})\big) \operatorname{diam}_{d_\eta} C. \tag{28}$$

Identifying C with a contracted set is a separate geometric step.

6.3. **Normal representations and moving centers.**

Lemma 6.3. *If the complex zero fiber contains a module-simple representation, the full normal Ext-quiver at every point of every M_η also has a module-simple representation in its complex zero fiber.*

Proof. The common module-simple locus gives a dense open subset of the complex quotient, hence of M_η by Corollary 4.3. Choose balanced representatives $x_j \to x_*$ from it. The common-parameter inverse in Theorem 4.4, at fixed parameter η, gives $w_j \to 0$ in the full normal W with both normal moments zero. The full equivariant slice and its complex corrections identify stabilizers up to conjugacy. Thus H_{w_j} is exactly the single common scalar circle, since this is the stabilizer of x_j. Proposition 4.1 complexifies it. At zero level, that proposition decomposes the representation into module-simple summands: zero-stability means that there are no proper subrepresentations. More than one summand or a repeated summand would give a nonscalar endomorphism. Hence w_j is module-simple. □

Apply all quiver statements to the full W, not to W' in (20). Properness and parameter convergence for $W'///_\nu H$ follow from the exact product and restriction to points with U-coordinate zero.

Theorem 6.4. *Let $\eta_n \to \eta_*$ and $p_n \in M_{\eta_n}$ approach $[x_*]$. In the common slice of Theorem 4.4, write their balanced normal coordinates as $(u_n, z_n) \to 0$ in $U \oplus W'$. Set*

$$\zeta_{H,n} = \eta_n|_{\mathfrak{h}} = (\eta_n - \eta_*)|_{\mathfrak{h}}.$$

For any $\sigma_n \downarrow 0$ such that

$$\zeta_{H,n}/\sigma_n^2 \to \nu, \qquad z_n/\sigma_n \to z$$

after unitary adjustment, the complete pointed limit with distances d_{η_n}/σ_n and base p_n is

$$U \times (W'///_\nu H), \qquad \text{with base } (0,[z]).$$

The same conclusion holds inside the actual family of Theorem 6.2, at physical scale $\delta_n\sigma_n$.

Proof. The exact model coordinate change is

$$(u, z') \longmapsto (u_n + \sigma_n u, \sigma_n z').$$

Translation in U does not affect the moments. It identifies the rescaled normal model with $U \times (W'///_{\nu_n} H)$, where $\nu_n = \zeta_{H,n}/\sigma_n^2$. On a fixed bounded radial region its relative squared-metric error in the original finite space is

$$O_{x_*}(|\eta_n - \eta_*| + |u_n| + \sigma_n R) = o(1). \tag{29}$$

In particular this error is not divided by σ_n.

Choose a fixed radius ρ whose twice-sized sublevel lies in the slice. Compact sublevels and the coarse metric comparison give exit cost at least $(\rho - \|(u_n, z_n)\|)/\sqrt{2}$. After division by σ_n this tends to infinity. Every competing path of length $O(\sigma_n)$ remains in the chart, and the radius of its translated coordinate $(u - u_n, z')$ is bounded by $\|z_n\| + \sqrt{2}$ times its length, hence is $O(\sigma_n)$. Conversely, admissible normal-model near-minimizers on bounded rescaled balls stay in bounded translated radial regions. Equation (29) therefore compares both intrinsic infima relatively; the common inverse supplies full ball coverage. Lemma 6.3 and Theorem 5.1, applied to full W, give the whole complete product limit.

In the actual family the outer error is $O(|t_n| + \delta_n^\alpha R^\alpha)$ on a bounded region of the first rescaled model. The inner error is (29). Both are multiplicative, so they add to first order and are not amplified by the smaller scale. The outer chart and the fixed inner slice each have diverging exit cost after rescaling by $\delta_n\sigma_n$. The same inverse-coverage and path argument therefore proves the actual conclusion. □

Tangential motion u_n may be arbitrarily large compared with σ_n while tending to zero. There is no comparability assumption between independent stability parameters. For example, when nonzero, $\sigma_n = \max(\|z_n\|, \sqrt{\|\zeta_{H,n}\|})$ gives bounded rescaled normal parameters and representatives, hence convergent subsequences. Smaller scales are allowed whenever the stated boundedness conditions hold.

Corollary 6.5. *Theorem 6.4 can be iterated through any fixed finite succession of convergent centers and positive scales tending to zero. At each stage require convergence of the retained parameter divided by the square of its scale and of the rescaled normal base representatives. The physical scale is the product of all stage scales, and the complete limit is the corresponding full normal quotient with its trivial Euclidean factors.*

Proof. Lemma 6.3 preserves the existence of a module-simple point in the complex zero fiber at every stage. Apply the preceding theorem at each fixed limiting center, with that center's own constants and radius. For finitely many stages the relative errors sum to $o(1)$; successive first-exit bounds and inverse charts give full ball coverage. No uniform radius over all possible centers or convergence of an infinite tower is asserted. □

6.4. **The size of the affine contracted fiber.** Let C_η be the fiber over the affine vertex under $Q_F \to Q_0$. It is nonempty and compact by the surjective projective face map.

Corollary 6.6. *There is C such that $\operatorname{diam}_{d_\eta} C_\eta \le C\sqrt{\|\eta\|}$ for every central η. The same upper bound holds for its actual image $f_t(C_{\zeta(t)})$ when $\zeta(t) \to 0$ and $t \to 0$.*

Proof. Lemma 4.2 bounds representative radii uniformly on $\|\eta\| = 1$. If the corresponding intrinsic diameters were unbounded, choose endpoints and parameters, pass to convergent representatives, and apply (21), a contradiction. Homogeneity gives the asserted scale, including the zero fiber, which is the vertex. The actual assertion follows by localization and (27). □

There is no uniform lower diameter estimate in this argument; secondary contractions can occur on smaller scales. The actual statement concerns the displayed image of the finite core. A global contraction is not identified by this estimate alone.

6.5. **Matching to the surrounding zero-level metric.** At a zero-level point x with scalar stabilizer, let $\lambda(x) > 0$ be the least eigenvalue of $C_x^* C_x$ on the total-trace-zero Hermitian vertex algebra, using its Hilbert–Schmidt norm.

Proposition 6.7. *If $\|\zeta\|/\lambda(x)$ is sufficiently small, the same complex orbit has a ζ-balanced representative $e^S x$ with*

$$\|S\|_{\mathrm{op}} \le A\frac{\|\zeta\|}{\lambda(x)}.$$

On the corresponding free quotient loci,

$$g_\zeta = \left(1 + O\left(\frac{\|\zeta\|}{\lambda(x)}\right)\right) g_0$$

as two-sided Hermitian-form inequalities. The relative comparison with the actual metric has error $O(|t| + \|x\|^\alpha + \|\zeta\|/\lambda(x))$.

Proof. Use the degree homotopy $\mu(e^S x) - s\zeta$ on the operator-norm ball $a = A\|\zeta\|/\lambda(x)$. For a strict spectral cut P put $P_0 = P - (\operatorname{tr} P/M)I$. Then

$$\|P_0\|^2 = \frac{k(M-k)}{M} \ge \frac{M-1}{M}, \qquad q_P(x) \ge \lambda(x)\|P_0\|^2.$$

The boundary estimate (3) is at least $2e^{-4a}a\lambda(x)\|P_0\|^2/(M-1)$, whereas the target pairing has absolute value at most $\|\zeta\|\|P_0\|$. Choose A large depending only on M and then a small. The homotopy has no boundary zeros and initial degree $+1$, giving the correction. No stronger inverse-function condition involving $\|\zeta\|\|x\|^2/\lambda(x)^2$ is used.

A zero-balanced point with scalar stabilizer is module-simple: (16) makes any invariant subspace reducing and hence produces a nonscalar stabilizer if proper. Thus these orbits belong to the common stable open and determine the same affine quotient point. On local holomorphic quotient tangents the derivative of the complex correction is vertical. Removing it by the full complex orbit infimum leaves only the action of e^S on ambient norms, whose squared distortion is $e^{\pm 4\|S\|_{\mathrm{op}}}$. This proves the finite metric estimate. Compose with (27) for the actual estimate. □

The matching statement applies when $\|\zeta\|/\lambda(x) \to 0$. The complete normal models above describe regimes where this outer condition fails. None of these local stationary results implies bubbling control or an HYM evolution theorem.

7. A rank-five K3 wall crossing

We construct one bundle-moduli problem to which all the preceding analytic results apply. It has three distinct stable factors, multiplicities $(2,2,1)$ and two independent polarization parameters. The two noncommutative multiplicity groups act on a common extension space. The three-edge cycle also removes the sign symmetry that would give the quadratic error automatically.

7.1. **The surface and the polystable bundle.** Consider the lattice with basis H, a, b and Gram matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -4 & 2 \\ 0 & 2 & -4 \end{pmatrix}. \tag{30}$$

In three hyperbolic planes with bases e_i, f_i, $e_i \cdot f_i = 1$, the vectors

$$H = e_0 + f_0, \qquad a = e_1 - 2f_1, \qquad b = 2f_1 + e_2 - 2f_2$$

have these pairings. Their e_0, e_1, e_2 coordinates form an identity minor, so the embedding is primitive. The lattice is even of signature $(1,2)$. Morrison's realization theorem [Mor84, Corollary 1.9(i)] gives a projective K3 surface with exactly this Néron–Severi lattice.

The class H can be made ample by choosing the marking. Indeed its orthogonal lattice has square $-4(p^2+q^2-pq)$ and contains no class of square -2. After choosing the positive-cone component, H lies in the interior of a root chamber. The root-reflection group carries it to the ample chamber; compose the marking with this action [Dol96, §1, Theorem 3.1 and Corollary 3.2]. This does not assert that the full Picard lattice is root-free: $H+a$ has square -2.

Let L_1, L_2, L_3 be the line bundles with classes $a, b, -a-b$. Their squares are -4 and their H-degrees are zero. Pairwise differences have square -12 and no sections in either sign: a nonzero effective divisor has positive H-degree, and the line bundles are distinct. Serre duality and Riemann–Roch give

$$\dim \operatorname{Ext}^1(L_i, L_j) = 4 \quad (i \neq j), \qquad \dim \operatorname{Ext}^\bullet(L_i, L_i) = (1,0,1). \tag{31}$$

Moreover $\chi(L_i(nH)) = 2 + \frac{1}{2}(c_1(L_i) + nH)^2 = n^2$. Thus

$$E_0 = L_1^{\oplus 2} \oplus L_2^{\oplus 2} \oplus L_3$$

is both slope and Gieseker polystable at H. Its invariants are

$$\operatorname{rk} E_0 = 5, \qquad c_1(E_0) = a+b, \qquad c_2(E_0) = 8, \qquad v(E_0) = (5, a+b, -5), \qquad v(E_0)^2 = 46. \tag{32}$$

The Mukai vector is primitive. The extension space has complex dimension 64, and the effective group $(GL_2 \times GL_2 \times GL_1)/\mathbb{C}^*$ has dimension 8.

For small real (t,u) the class $H + ta + ub$ is Kähler. Choose its smoothly varying Ricci-flat forms. The real moment parameter in (5) is

$$\zeta(t,u) = \frac{2\pi}{5}(18t - 12u,\ -12t + 18u,\ -12t - 12u). \tag{33}$$

Its image is the entire plane $2\zeta_1 + 2\zeta_2 + \zeta_3 = 0$. Thus the analytic family realizes every sufficiently small real central parameter, not just one prescribed ray.

7.2. **Simple zero-level representations.** Orient the edges $12, 13, 23$ toward the larger index. On each edge take four forward arrows and four reverse arrows. For rectangular matrices C, D put

$$(A_1, A_2, A_3, A_4) = (C, C, D, D), \qquad (B_1, B_2, B_3, B_4) = (C^*, -C^*, D^*, -D^*). \tag{34}$$

The complex moment cancels between the two copies of each matrix; each forward/reverse pair cancels in the real moment. For dimensions $(2, 2, 1)$ use

$$(C_{12}, D_{12}) = \left(I_2, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right), \qquad (C_{13}, D_{13}) = (C_{23}, D_{23}) = \big((1, 0), (0, 1)\big).$$

Lemma 7.1. *These arrows give a module-simple point in the complex and real zero moment fiber. The same construction gives simple zero-level representations for every summand in Table 1.*

Proof. An endomorphism commutes with I_2 on edge 12, so its first two vertex matrices agree. That matrix commutes with the displayed nilpotent and its adjoint, hence is scalar. The coordinate rows make its scalar agree with the scalar at vertex 3. Thus its endomorphism algebra is $\mathbb{C}$. The directed arrow collection is closed under adjoints up to sign, so any subrepresentation has an invariant orthogonal complement. Its orthogonal projection would be a nonscalar endomorphism if the subrepresentation were nonzero and proper. This proves module simplicity, rather than only the Schur property.

For the other vectors, on edges of sizes $(1, 2)$ use the two coordinate columns, on $(2, 1)$ the two coordinate rows, and on $(1, 1)$ use $C = 1, D = 0$. Use the same (I_2, D) at a $(2, 2)$ edge when present. These arrows and their adjoints force scalar endomorphisms at every connected vertex and the scalars agree. Every listed support is connected; a singleton has dimension one and is already simple. The same reducing-subspace proof applies. □

At any of these fixed simple zero-level points the real orbit derivative is positive definite modulo scalars. The implicit-function theorem therefore solves all sufficiently small supported central levels in the same complex orbit. Scaling arrows by δ scales both moments by δ^2. Choosing $\delta = C\sqrt{\|\zeta\|}$ gives representatives of norm $O(\sqrt{\|\zeta\|})$ in every sufficiently small direction. At level zero, scaling a nonzero simple point gives arbitrarily small simple points. A vertex-simple factor is represented by zero arrows and only the zero supported level.

In particular the local actual moduli contain stable bundles. At a stable point their complex dimension is $v^2 + 2 = 48$, equivalently $64 - 2 \cdot 8$. Here nonemptiness comes from the explicit construction and Theorem 3.1, not from dimension subtraction. Lemma 7.1 also verifies the hypothesis of Theorem 5.1.

7.3. **Five walls and their actual strata.** For $0 \le d \le (2, 2, 1)$ the equation $d \cdot \zeta = 0$ is

$$(d_1 - 2d_3)\zeta_1 + (d_2 - 2d_3)\zeta_2 = 0.$$

For $d_3 = 0$ the coefficients range from 0 to 2; $d_3 = 1$ gives complementary negatives. Omitting the zero and full vectors and removing proportional equations leaves exactly

$$\zeta_1 = 0, \quad \zeta_2 = 0, \quad \zeta_1 + \zeta_2 = 0, \quad \zeta_1 + 2\zeta_2 = 0, \quad 2\zeta_1 + \zeta_2 = 0. \tag{35}$$

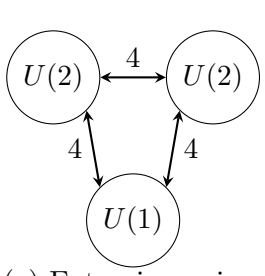
They divide the plane into ten chambers (Figure 1). Off these lines, the trace inequality (16) is strict on every proper nonzero invariant subspace. Thus balanced representations are stable and have scalar stabilizer.

This last statement does not mean module simplicity. For example take all reverse arrows zero, the four matrix units on edge 12, and the two coordinate rows on edges 13 and 23,

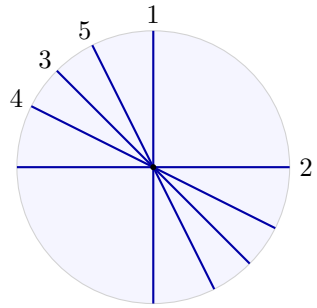
with the other slots zero. The complex moment is zero, the real level is $(-3, 1, 4)$, and the endomorphism algebra is scalar. That level lies off all five walls, but the third vertex alone is a proper subrepresentation. Module simplicity was used above only for the explicitly adjoint-closed zero-level representations.

TABLE 1. Realized walls and the selected two-factor strata. Here $e = (2, 2, 1) - d$, $k = \dim \mathrm{Ext}^1(F_d, F_e)$, and $U = \mathrm{Ext}^1(F_d, F_d) \oplus \mathrm{Ext}^1(F_e, F_e)$.

No.	Normal	d	e	k	$\dim_{\mathbb{C}} U$	Transverse dimension
1	$(1, 0)$	$(1, 0, 0)$	$(1, 2, 1)$	10	30	18
2	$(0, 1)$	$(0, 1, 0)$	$(2, 1, 1)$	10	30	18
3	$(1, 1)$	$(1, 1, 0)$	$(1, 1, 1)$	12	26	22
4	$(1, 2)$	$(1, 2, 0)$	$(1, 0, 1)$	18	14	34
5	$(2, 1)$	$(2, 1, 0)$	$(0, 1, 1)$	18	14	34



(a) Extension quiver



(b) Five walls, ten chambers

FIGURE 1. The rank-five example. Each edge in (a) denotes four arrows in each direction; the diagonal circle acts trivially. In (b) the horizontal and vertical coordinates are ζ_1, ζ_2. Wall numbers refer to Table 1; the bounding circle has no geometric significance.

For each row take the simple zero-level representations for d, e and scale them into the original bundle chart. Equivariance preserves their block decomposition. Theorem 3.1 and the local morphism identification produce locally free H-stable bundles F_d, F_e: their solutions are polystable, and scalar endomorphisms imply stability. They have

$$v_d = \sum_i d_i v(L_i), \qquad v_e = \sum_i e_i v(L_i), \qquad \chi(F_d) = \chi(F_e) = 0.$$

Their H-slopes vanish, so their reduced H-Hilbert polynomials are equal. Their distinct Mukai vectors make them nonisomorphic.

Along the corresponding line $d \cdot \zeta = 0$ their slopes remain equal. The fixed simple-orbit corrections above balance each factor at all sufficiently small levels on this line, in the same holomorphic isomorphism classes. Hence these factors are stable there and their sum is polystable. This verifies both the H and the recentered real-class hypotheses of Theorem 3.1 at every selected point. We do not assert that all factors at every nearby wall point are H-stable.

For dimension vectors d, e the Mukai pairing is

$$(v_d, v_e) = -2\sum_i d_i e_i + 4\sum_{i<j}(d_i e_j + d_j e_i). \tag{36}$$

At these distinct equal-slope stable bundles, Hom in either direction vanishes; Serre duality and Riemann–Roch identify (36) with the cross-extension dimension k. Their self-extension dimensions are $v_d^2 + 2, v_e^2 + 2$. This gives Table 1, with $\dim U = 50 - 2k$ and transverse dimension $2(k-1)$, summing to 48.

7.4. **Transverse contractions and metric profiles.** At a fixed selected wall point, the full local representation is

$$U \oplus \mathbb{C}^k \oplus (\mathbb{C}^k)^*,$$

with an effective circle trivial on U and of opposite weights on the two remaining factors. Write them as (A, B). The equations and complex action are

$$BA = 0, \qquad \|A\|^2 - \|B\|^2 = \kappa, \qquad z \cdot (A, B) = (zA, z^{-1}B).$$

At $\kappa > 0$, $A \neq 0$ determines a line, and $BA = 0$ identifies the quotient with $T^*\mathbb{P}^{k-1}$. The negative sign gives the dual projective description. At zero the invariant $N = AB$ satisfies $N^2 = 0$, $\operatorname{rk} N \leq 1$, and gives the corresponding affine cone. The two incidence maps are projective, isomorphisms over $N \neq 0$, with zero fiber $\mathbb{P}^{k-1}$.

These finite contractions transport to actual local analytic contractions on saturated neighborhoods. The full central analytic germ is supplied by Theorem 3.1; the generic source charts are smooth and holomorphically identified. For a fixed small κ, the compact exceptional fiber scales into radius $R/2$ of a common chart. Properness of the finite contraction makes the image of its closed complement $\{r \geq R\}$ closed and disjoint from the target origin. Remove this image and shrink inside the central target chart. The resulting neighborhood U_κ has its entire inverse image in the source chart. The contraction is therefore defined on a full saturated neighborhood and remains proper and projective after analytic transport. Its target neighborhood may depend on κ; no uniform saturated radius is required.

Theorem 7.2. *The rank-five K3 family constructed above has the following properties for its actual harmonic L^2 metric.*

1. *Near the three-factor point its full local moduli metrics have relative error $O(|t| + |u| + r)$ from the doubled-triangle quotients, uniformly through all stabilizer changes. Every convergent rescaled parameter direction gives the complete quotient limit of Theorem 6.2, including the five singular wall directions and their intersection. The finite iterated moving-center limits of Corollary 6.5 apply.*
2. *Each wall in (35) has selected actual two-factor strata with the dimensions in Table 1. Fix such a point and let $\kappa \to 0$ with constant sign ϵ. For any sequence of base points whose balanced representatives divided by $\sqrt{|\kappa|}$ converge to $(v_*, [A_*, B_*])$, the actual distances divided by $\sqrt{|\kappa|}$ converge in the pointed Gromov–Hausdorff sense to the complete product $U \times T^*\mathbb{P}^{k-1}$, based at that limit. The transverse factor carries its flat quotient metric at level $\epsilon \in \{1, -1\}$; for negative level the projective space has the dual description. The actual projective exceptional fiber has the precise $\sqrt{|\kappa|}$ diameter profile (28).*
3. *At zero transverse parameter, let s be the nonzero singular value of $N = AB$ and let $\mathcal{S}$ denote the local two-factor stratum. In its recentered coordinates,*

$$d^{L^2}(f_0(v, N), \mathcal{S}) = (1 + O(r))\sqrt{2s}.$$

At $N = 0$ both sides vanish.

(4) *On the free surrounding zero-level locus, Proposition 6.7 gives relative matching error $O(|t| + |u| + r + \|\zeta\|/\lambda)$ whenever $\|\zeta\|/\lambda \to 0$.*

Proof. All geometric hypotheses of Theorem 3.1 and the existence of a module-simple point in the complex zero fiber are verified above. Apply Theorems 6.2 and 6.4 for the first assertion. At a fixed selected wall point, the exact product normal description and $\sqrt{|\kappa|}$ homogeneity give the second assertion, including coverage of the whole complete product. The saturated-domain construction identifies its compact projective core with the actual contracted fiber.

At zero transverse level, balance gives $\|A\| = \|B\|$. The unique nonzero singular value of AB is $s = \|A\|\|B\|$, so transverse distance to the cone vertex is $\sqrt{\|A\|^2 + \|B\|^2} = \sqrt{2s}$. The Euclidean U factor can be held fixed and cannot reduce this transverse distance. The two-factor stratum corresponds exactly to $U \times \{0\}$: its nearby summands stay stable and nonisomorphic and equivariance preserves their decomposition. Proposition 6.1 gives the third assertion, even when the transverse distance is much smaller than r. The fourth is the previously proved outer matching theorem. □

A vertex-scalar unitary negating every arrow of a triangle would have scalar ratios -1 on each edge, whose product would give $1 = -1$. Thus the sufficient parity hypothesis does not apply here. We claim the proved linear spatial error, without ruling out additional cancellations under other normalizations.

The explicit matrices, lattice embedding and pairing formulas above make the finite data reproducible. An accompanying exact rational calculation checks all eleven simple representations, their real and complex moment equations, centralizer dimensions, and the five wall normals.[1] The analytic conclusions use the proofs above, not numerical asymptotics. All contractions and metric limits in this application are local bundle-moduli statements; no global Gieseker wall-crossing identification or bubbling compactification is inferred.

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